

MEMORY

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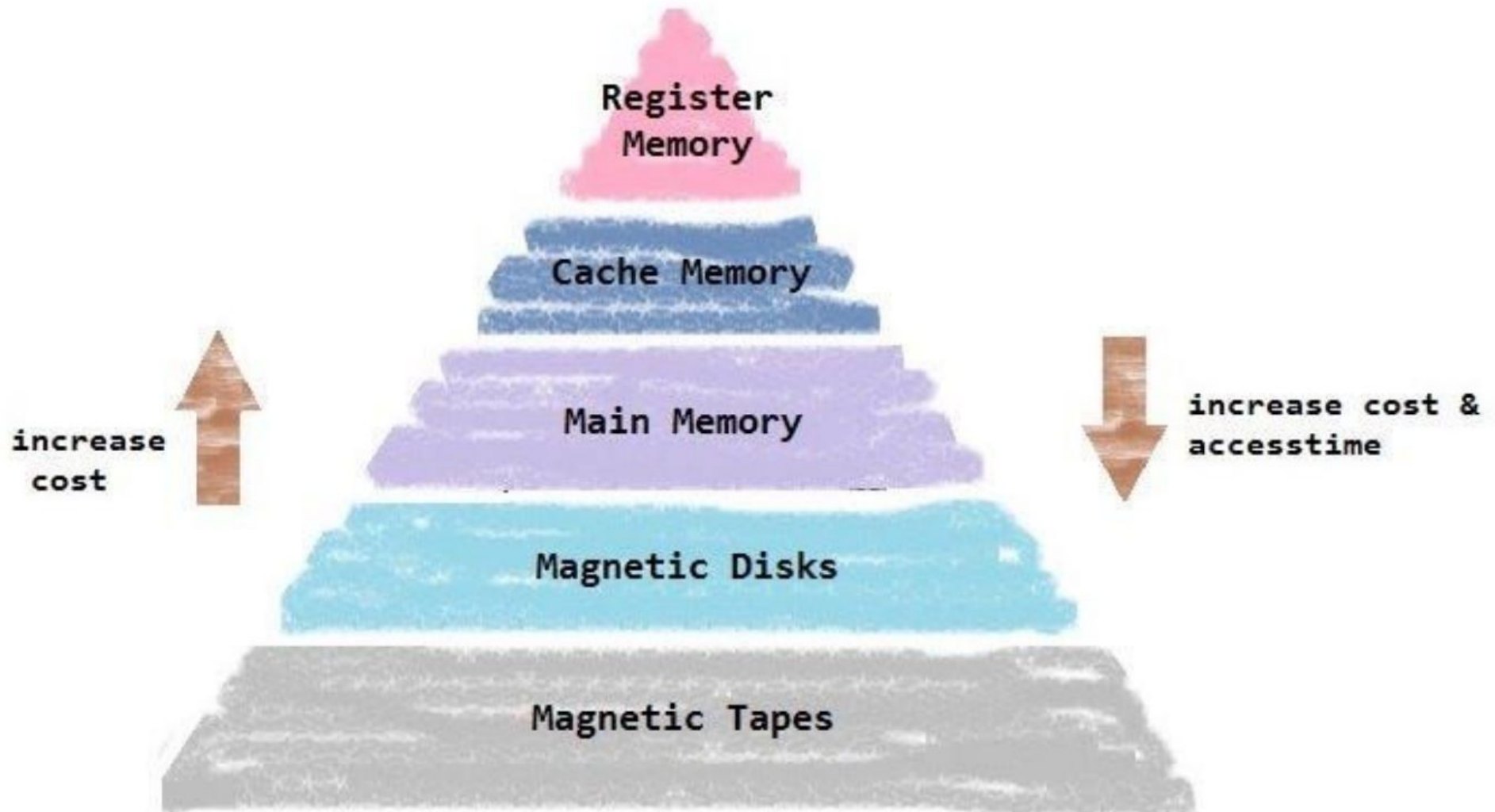
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What is Computer Memory

“Memory” means the place where we have to **store** any thing.

Memory is also very important for computer system because in computer system we have to store some data or information and for storing these items we need some memory or space.

Memory Hierarchy

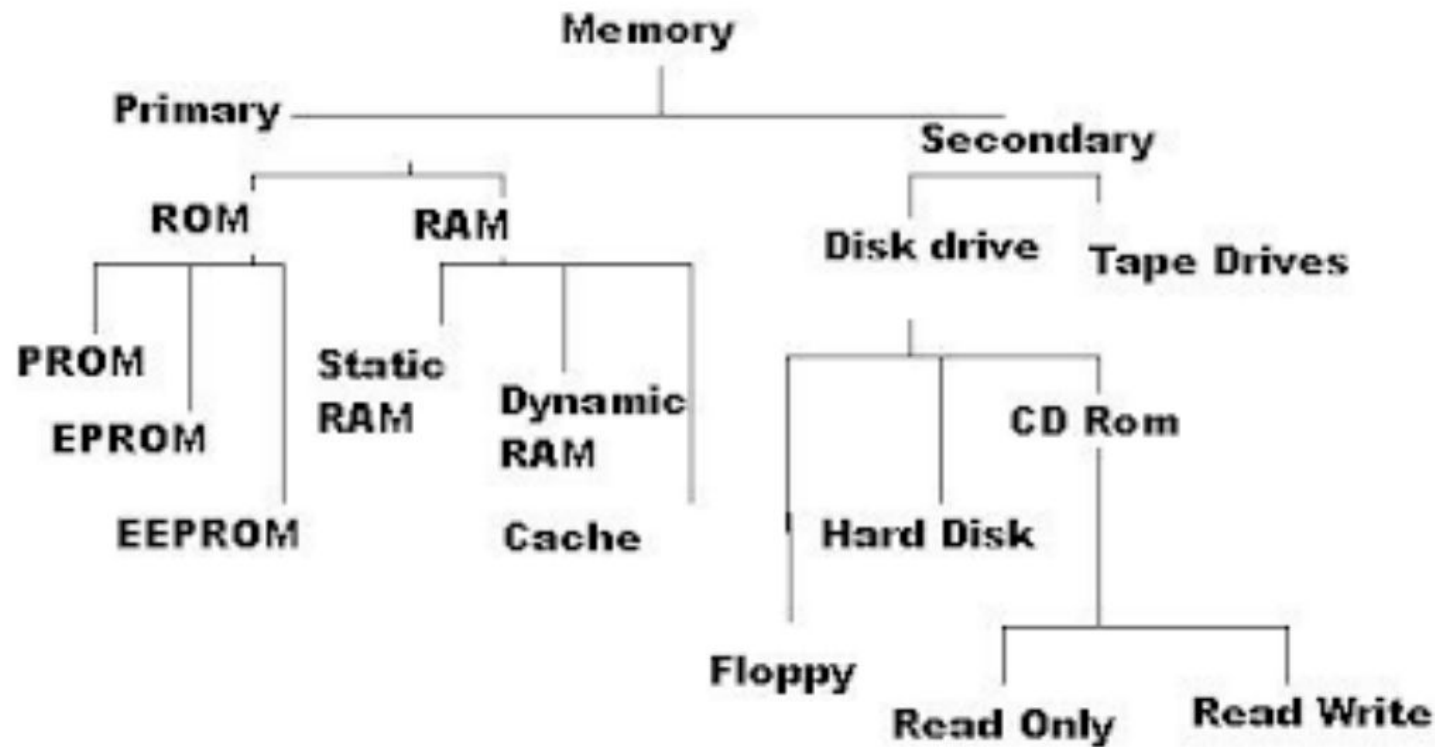


Units of Memory

There are various units which are used to measure computer memory:

- **Bit**- Smallest unit of computer memory
- **Byte** - 8 bit = 1 byte
- **Kilobyte** - 1024 byte = 1 kb
- **Megabyte** - 1024 kb = 1 mb
- **Gigabyte** - 1024 mb = 1 gb
- **Terabyte** - 1024 gb = 1 tb

Classification of Memory



Primary Memory

Primary memory also known as "main memory" or "internal memory" which is located in the motherboard of system.

- Directly connected to the CPU
- Place where only little bit of data are stored

This is further divided into two parts :

- RAM
- ROM

RAM

RAM also known as "**random access memory**"

RAM is a circuit which the user currently uses and have its separate space or slot in motherboard.

- RAM stores the data temporary in it
- needs power or electricity to work
- when the computer power is turned off then all the data in it will be erased automatically
- allows the computer to run the software faster

RAM is further divided into two parts :

□ **SRAM**

□ **DRAM**

SRAM

SRAM known as **Static RAM**

SRAM is a type of random access memory that retains data bits in its memory as long as power is being supplied.

- SRAM does not need to be refreshed periodically
- there are so many programs which runs in background
- SRAM is faster than DRAM
- more expensive than DRAM

DRAM

DRAM known as **Dynamic RAM**

DRAM is a type of random access memory that contains data only when electricity is available.

- DRAM needs to be refreshed periodically
- DRAM is slower than SRAM
- cheaper than SRAM
- Mainly used in PC
- requires less space

ROM

ROM also known as "**read only memory**"

ROM is a memory that can only read the data which is stored in it.

- RAM is used when we turn on our computer
- Contains BIOS (basic input/output system) of system
- This is non-volatile in nature

Cache Hit/Miss

Access Time:

It is the time interval between the read/write request and the availability of the data. As going down the Hierarchy, the access time increases.

Hit

- A **hit** occurs when the CPU requests data, and the **data is found in the cache**.
- Since the cache is a smaller, faster memory located closer to the CPU, accessing data from it is much quicker than accessing it from the main memory.

Miss

- A **miss** occurs when the CPU requests data, but it is **not found** in the cache.
- When there's a miss, the data must be fetched from the slower main memory, which takes more time. After fetching from the main memory, the data is usually copied into the cache for faster access in the future.

Cache Hit/Miss (Cont...)

Hit Ratio:

The **hit ratio** is a measure of how often data is found in the cache, expressed as:

Hit Ratio (h)=Number of cache hits / Total number of memory accesses

- ★ A higher hit ratio indicates that the cache is working effectively, and most of the required data is found in the cache.

Miss ratio

The **miss ratio** is a measure of how often data is not found in the cache, expressed as: Miss Ratio (m)=1–Hit Ratio

Cache Hit/Miss (Cont...)

Suppose a CPU accesses memory 100 times, and out of those 100 times, the data is found in the cache 90 times.

This means:

- **Hit Ratio** = $90/100 = 0.9$ (90%)
- **Miss Ratio** = $1 - 0.9 = 0.1$ (10%)

Question: A computer system has a cache access time of 200 ns, a memory access time of 1500 ns, and 60 out of 100 memory accesses are cache hits; calculate the hit ratio, miss ratio, and the effective access time.

Given,

Cache access time = 200ns

Memory access time = 1500ns

Number of cache hits = 60

We know,

$$\begin{aligned}\text{hit ratio} &= \frac{\text{Number of cache hits}}{\text{Total number of memory accesses}} \\ &= \frac{60}{100} = 0.6\end{aligned}$$

$$\begin{aligned}\text{miss ratio} &= 1 - \text{hit ratio} \\ &= 1 - 0.6 \\ &= 0.4\end{aligned}$$

When any cache miss occurs, both the cache and main memory are accessed.

$$\begin{aligned}\text{So, access time for a miss} &= \text{Cache access time} + \text{memory access time} \\ &= 200\text{ns} + 1500\text{ns} \\ &= 1700\text{ns}.\end{aligned}$$

$$\begin{aligned}\text{now, Effective access time} &= (\text{hit ratio} \times \text{access time for a hit}) \\ &\quad + (\text{miss ratio} \times \text{access time for a miss}) \\ &= (0.6 \times 200\text{ns}) + (0.4 \times 1700\text{ns}) \\ &= 120\text{ns} + 680\text{ns} \\ &= 800\text{ns}\end{aligned}$$

Thank you